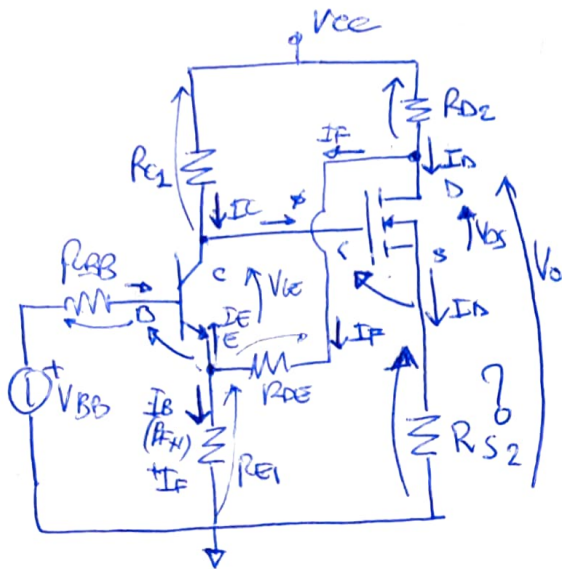


① all caps opened

1.1 $R_{S2} = ?$ $V_O = 5V$



$$V_{BB} = V_{CC} \cdot \frac{R_{B2}}{R_{B1} + R_{B2}} = 3.27V$$

$$R_{BB} = R_{B1} \parallel R_{B2} = 117.6k\Omega$$

222.2kΩ

~~$$I_B = \frac{V_{BB} - V_{BE}}{R_{BB}}$$~~

can't
FUCK!

$$\begin{cases} V_O = V_{CC} - R_{C2} \cdot (I_D + I_F) \\ V_{BB} = R_{BB} \cdot I_B + V_{BE} + R_{E1} \cdot (I_B(\beta_F + 1) + I_F) \\ \frac{V_O - R_{E1} \cdot (I_B(\beta_F + 1) + I_F)}{R_{DE}} = I_F R_{OE} \end{cases}$$

$$\begin{cases} I_F (R_{DE} + R_{E1}) = V_O - R_{E1} \cdot I_B \cdot (\beta_F + 1) \rightarrow I_F = \frac{V_O - R_{E1} \cdot I_B \cdot (\beta_F + 1)}{R_{DE} + R_{E1}} \\ V_{BB} - V_{BE} = R_{BB} \cdot I_B + R_{E1}(\beta_F + 1) \cdot I_B + R_{E1} \cdot \frac{V_O - R_{E1} \cdot I_B \cdot (\beta_F + 1)}{R_{DE} + R_{E1}} \\ -V_O = -V_{CC} + R_{C2}(I_D + I_F) \rightarrow I_D = \frac{V_{CC} - V_O}{R_{C2}} - I_F \end{cases}$$

$$\begin{cases} I_F = \frac{V_O - R_{E1} \cdot I_B \cdot (\beta_F + 1)}{R_{DE} + R_{E1}} \\ V_{BB} - V_{BE} - V_O \cdot \frac{R_{E1}}{R_{DE} + R_{E1}} = I_B \left[R_{BB} + R_{E1}(\beta_F + 1) - \frac{R_{E1}^2}{R_{DE} + R_{E1}}(\beta_F + 1) \right] \\ R_{C2} = \frac{V_{CC} - V_O}{I_D + I_F} \end{cases}$$

$$I_B = \frac{V_{BB} - V_{BE} - V_O \cdot \frac{R_{E1}}{R_{DE} + R_{E1}}}{\frac{R_{BB}(R_{DE} + R_{E1}) + (\beta_F + 1)[R_{E1} \cdot R_{DE}]}{R_{DE} + R_{E1}}} = \frac{(V_{BB} - V_{BE}) \cdot \frac{R_{E1}}{R_{DE} + R_{E1}} - V_O R_{E1}}{R_{BB}(R_{DE} + R_{E1}) + R_{E1} \cdot R_{DE}(\beta_F + 1)}$$

$I_F = 0.297 \mu A$ $0.328 \mu A$

$R_{C2} \rightarrow I_D = 0.438 \mu A$ $0.407 \mu A$

$= 7.13 \mu A$ $5.3 \mu A$

$I_C = 1.06 \mu A$

Let's solve I_D

$$\begin{cases} V_{GS} = -V_{CE} + R_{S1} I_C + R_{S2} I_D \\ I_D = I_{DSS} \cdot \left(\frac{V_{GS}}{V_T} - 1 \right)^2 \end{cases} \quad \begin{cases} R_{S2} = \frac{V_{CE} - V_{GS} - R_{S1} I_C}{I_D} \\ \frac{I_D}{I_{DSS}} = \frac{V_{GS}^2}{V_T^2} + 1 - 2 \frac{V_{GS}}{V_T} \end{cases}$$

$$V_{GS}^2 \cdot \frac{1}{V_T^2} + V_{GS} \cdot \frac{2}{V_T} + 1 - \frac{I_D}{I_{DSS}} = 0 \rightarrow V_{GS} = \frac{3.32V}{3.27}$$

$$R_{S2} = \boxed{6.5k\Omega} \quad 7.94k\Omega$$

1.2 OPs

$$V_{CE} = V_{CC} - R_{C1} I_C - R_{E1} (I_F + I_B \cdot (\beta_F + 1)) = \frac{3.55V}{5.12V}$$

$$I_C = \beta_F I_B = \frac{1.43mA}{1.06mA}$$

$$Q_1: (\frac{3.55}{5.12}, \frac{1.43}{1.06mA}) [V, A] \quad (5.12, 1.06mA) [V, A]$$

$$V_{DS} = V_0 - R_{S2} I_D = \frac{3.03V}{1.77V}$$

$$Q_2: (\frac{3.03}{1.77}, \frac{0.438mA}{0.407mA}) [V, A] \quad (1.77, 0.407mA) [V, A]$$

1.3 A_v

output: voltage sampling
input: voltage mixing } voltage amplifier



$$R_{if} = R_{E1} // R_{DE} = 0.92k\Omega$$

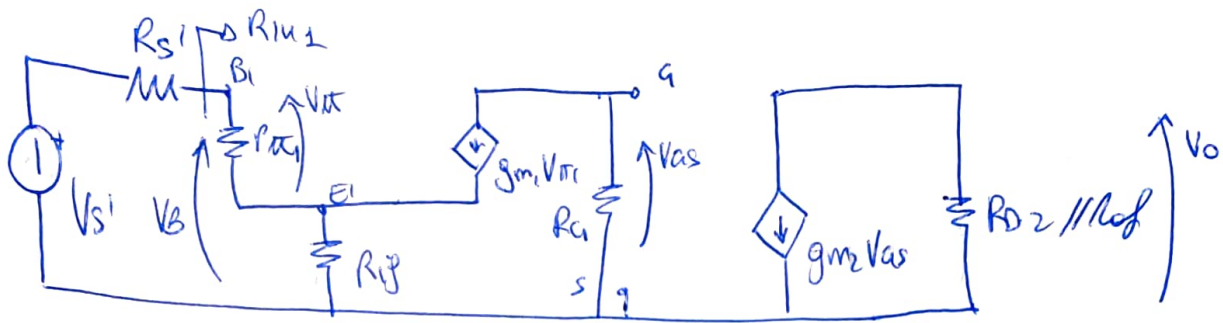
$$R_{of} = R_{DE} + R_{E1} = 12k\Omega$$

$$\beta = \frac{R_{E1}}{R_{DE} + R_{E1}} = 1/12$$

Eq. Voltage source performing theorem:

$$V_S' = V_S \cdot \frac{R_{BB}}{R_{BB} + R_S}$$

$$R_S' = R_{BB} // R_S = 0.992k\Omega$$



$$R_{in1} = r_{\pi 1} + R_{iB} \cdot (\beta_{01} + 1) = 188.6 \text{ k}\Omega \quad 189.8 \text{ k}\Omega$$

$$g_{m1} = \frac{I_C}{V_T} = 55 \text{ mS} \quad 40.7 \text{ mS}$$

$$A^{\alpha} = \frac{V_o}{V_{s'}} = \frac{V_o}{V_{B1}} \cdot \frac{V_{B1}}{V_B} \cdot \frac{V_B}{V_{s'}}$$

$$V_{B1} = \frac{\beta_{01}}{g_{m1}} = 3.66 \text{ k}\Omega \quad 4.905 \text{ k}\Omega$$

$$\frac{V_o}{V_{B1}} = -g_{m2} \cdot R_{D2} \parallel R_{o2} = -2.88 \text{ V/V} \quad -2.70 \text{ V/V}$$

$$g_{m2} = \frac{2I_D}{V_{GS} - V_T} = 0.664 \text{ mS} \quad 0.640 \text{ mS}$$

$$\frac{V_{B1}}{V_B} = -\frac{\beta_{01} \cdot R_{C1}}{R_{in1}} = -3.5 \text{ V/V}$$

$$\frac{V_B}{V_{s'}} = \frac{R_{in1}}{R_{in1} + R_{s'}} = 0.995 \text{ V/V}$$

$$\rightarrow A^{\alpha} = 10.03 \text{ V/V} \quad 9.40$$

$$A^F = \frac{A^{\alpha}}{1 + \beta A^{\alpha}} = 5.66 \text{ V/V} \quad 5.27 \text{ V/V}$$

$$A = \frac{V_o}{V_s} = \frac{V_o}{V_{s'}} \cdot \frac{V_{s'}}{V_s} = A^F \cdot \frac{R_{BB}}{R_{BB} + R_s} = \boxed{5.62 \text{ V/V}} \quad 5.25 \text{ V/V}$$

1.4 R_{in}

$$R_{inOL} = R_{s'} + R_{in1} = 189.6 \text{ k}\Omega \quad 190.8$$

$$R_{inF} = R_{inOL} \cdot (1 + \beta A^{\alpha}) = 347.6 \text{ k}\Omega \quad 340.2 \text{ k}\Omega$$

$$R_{in} = \frac{R_{inF} \cdot R_{BB}}{R_{inF} + R_{BB}} = 88.8 \text{ k}\Omega \quad 135.3 \text{ k}\Omega \quad 16.2$$

1.5 f_L

$$f_L = \frac{1}{2\pi} \cdot \left(\frac{1}{C_{s2} \cdot R_{s2}^{sc}} + \frac{1}{C_s \cdot R_s^{sc}} \right) = \frac{1}{2\pi} \left(\frac{1}{112.6} + \frac{1}{48.3} \right) = \boxed{25 \text{ kHz}} \quad 18.6 \text{ kHz}$$

$$R_{s2}^{sc} = R_s + (R_{in} - R_s) = R_{in} = 88.8 \text{ k}\Omega \quad 135.3 \text{ k}\Omega$$

usually capacitors on the emitter/source ~~are not~~ follow the rule $\frac{F\alpha}{1 + \beta A^{\alpha}}$ for f_L therefore

$$R_{s2}^{sc} = R_{s2}^{OL} \cdot (1 + \beta A^{\alpha}) = (R_{s2} \parallel \frac{1}{g_{m2}}) \cdot (1 + \frac{10}{12}) = 2.07 \text{ k}\Omega \quad 2.33 \text{ k}\Omega$$

1.6 Part

$$R_{out}^{OL} = R_{out} // R_{of} = 4.34 \text{ k}\Omega$$

$$R_{out}^F = \frac{R_{out}^{OL}}{1 + \beta A_{OL}} = \frac{2.37 \text{ k}\Omega}{2.43 \text{ k}\Omega} = R_{out}$$

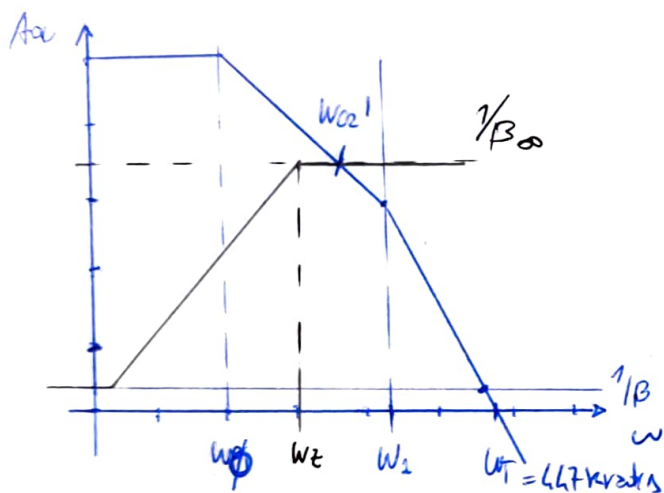
16.30

(2)

2.1. A_v ideal

$$A_v = 1 + \frac{R_2}{R_1} = \boxed{+3.06} \quad \checkmark$$

2.2 B + PM



$$\beta = \frac{R_1}{R_2 + R_1} \rightarrow \frac{1}{\beta} = \frac{R_2 + R_1}{R_1} = 3.06$$

$$A_v \cdot w_p = A_1 \cdot w_z \rightarrow A_1 = \frac{A_v \cdot w_z}{w_1}$$

$$A_1 \cdot w_z^2 = \frac{1}{\beta} \cdot w_{cr}^2 \rightarrow$$

$$w_{cr} = \sqrt{\beta \cdot A_v \cdot w_z \cdot w_1} = 31.6 \text{ krad/s}$$

$$B = w_{cr} = \boxed{311.6 \text{ krad/s}} \quad \checkmark$$

$$PM = 180 - \tan^{-1}\left(\frac{w_{cr}}{w_p}\right) - \tan^{-1}\left(\frac{w_{cr}}{w_1}\right) = \boxed{3.4^\circ} \quad 4.5^\circ \quad \checkmark$$

2.3

with compensated network we have:

$$\beta_{oo} = \frac{R_1 // R_c}{R_2 + R_1 // R_c} \approx \frac{R_c}{R_2 + R_c} = 3.23 \cdot 10^{-4} \rightarrow \frac{1}{\beta_{oo}} = 3092 \sim 70 \text{ dB}$$

$$w_z = \frac{1}{C R_c} = 1.38 \text{ krad/s} \quad \checkmark$$

$$w_p = \frac{1}{\beta_{oo} \cdot w_p} = \frac{1}{\beta_{oo}} \cdot \frac{1}{w_z} \rightarrow w_p = w_z \cdot \frac{\beta_{oo}}{\beta_{oo}} = 0.916 \text{ rad/s}$$

$$PM = 180 - \tan^{-1}\left(\frac{w_{cr}}{w_p}\right) - \tan^{-1}\left(\frac{w_{cr}}{w_1}\right) - \tan^{-1}\left(\frac{w_{cr}}{w_p}\right) + \tan^{-1}\left(\frac{w_{cr}}{w_z}\right) = \boxed{59.5^\circ}$$

$$w_{cr} = A_v \cdot w_z \cdot \beta_{oo} = 3.23 \text{ krad/s}$$

2.4 $A(s) = V_o/V_s$

$$A(s) = 1 + \frac{R_2}{R_1}$$